ME25102: Basic Thermodynamics

Fundamentals-2 Work and Heat

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Course Objectives						
1	To understand the fundamental concepts and laws of thermodynamics					
Course Outcomes						
1	**Describe the basics of thermodynamics with heat and work interactions.**					

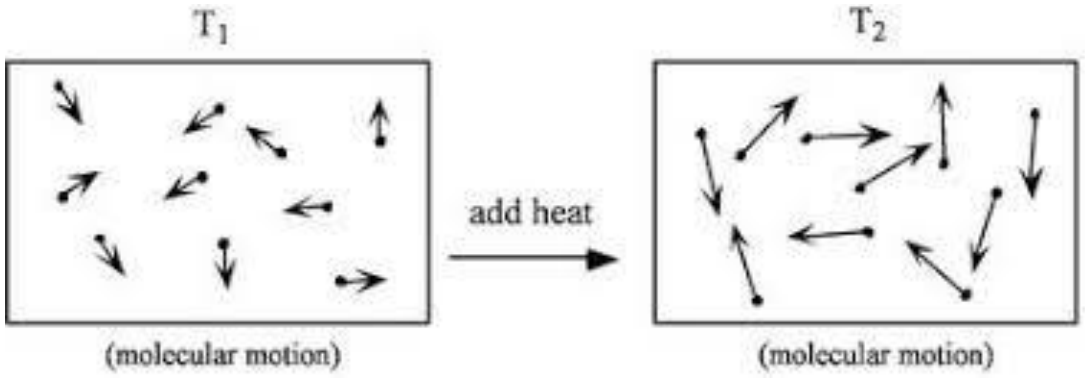
Definition of heat, Concept of heat and work, Sign convention and its conversion. Modes of heat transfer: conduction, convection and radiation, Fourier's law, Newton's law of cooling, Stefan Boltzmann's law. (Simple numerical)

Energy-

- The capacity to do work
  - Potential Energy, Kinetic Energy, Thermal, Electrical, Mechanical, Magnetic
  - Elastic, Chemical, Nuclear, Light, Sound
- Energy Stored- Chemical, Potential, Nuclear, Kinetic, Potential, Elastic etc
- Energy in Transit- Work and Heat

Heat-

- **Energy due to** the movement of atoms, molecules or ions in solids, liquids and gases.



- Unit- **cal or kcal**
- (Energy required to increase the temperature of 1kg water)
- SI Unit - **Joule (J) or kilo Joule (kJ)**

Work-

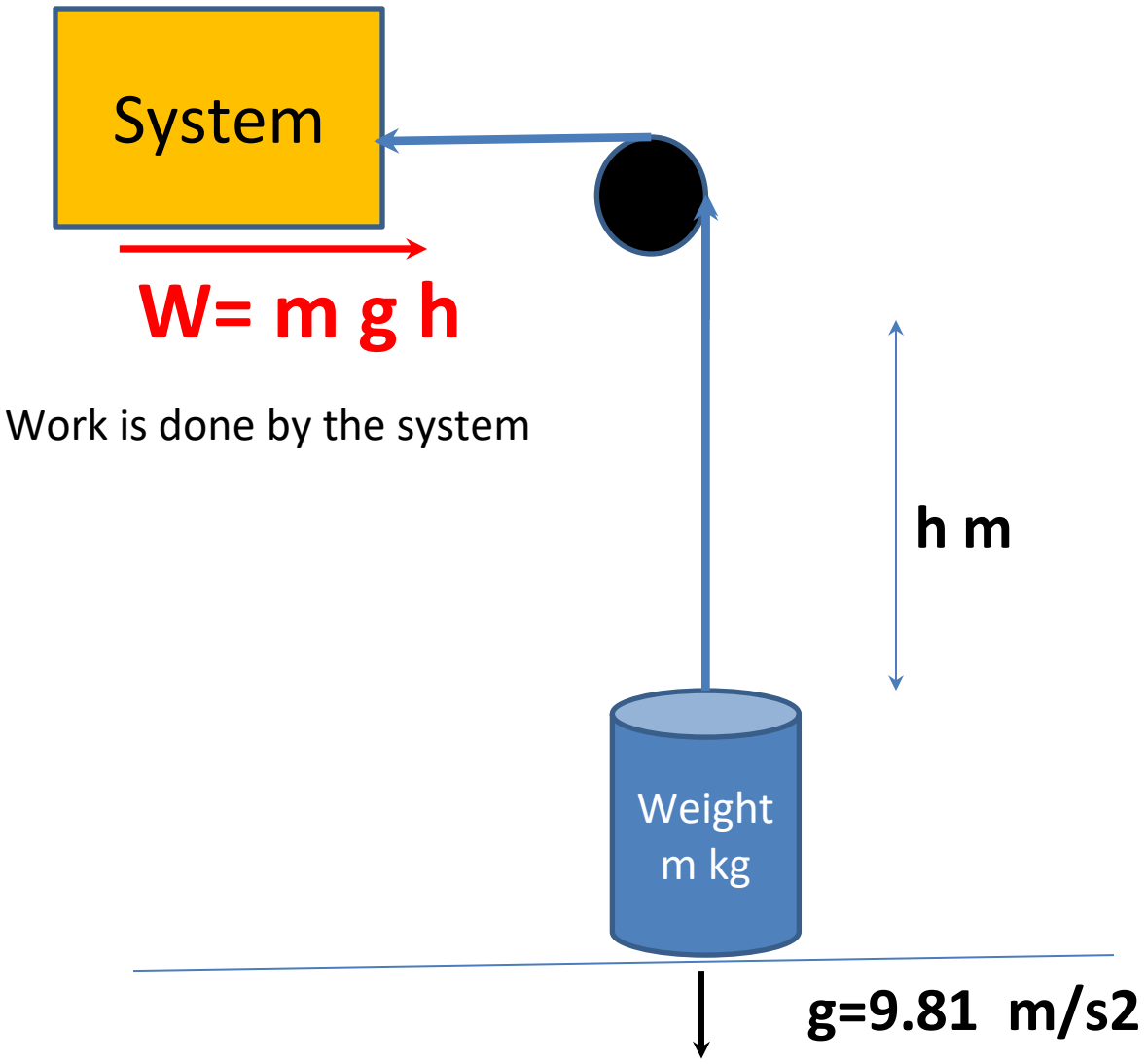
- Definition from physics

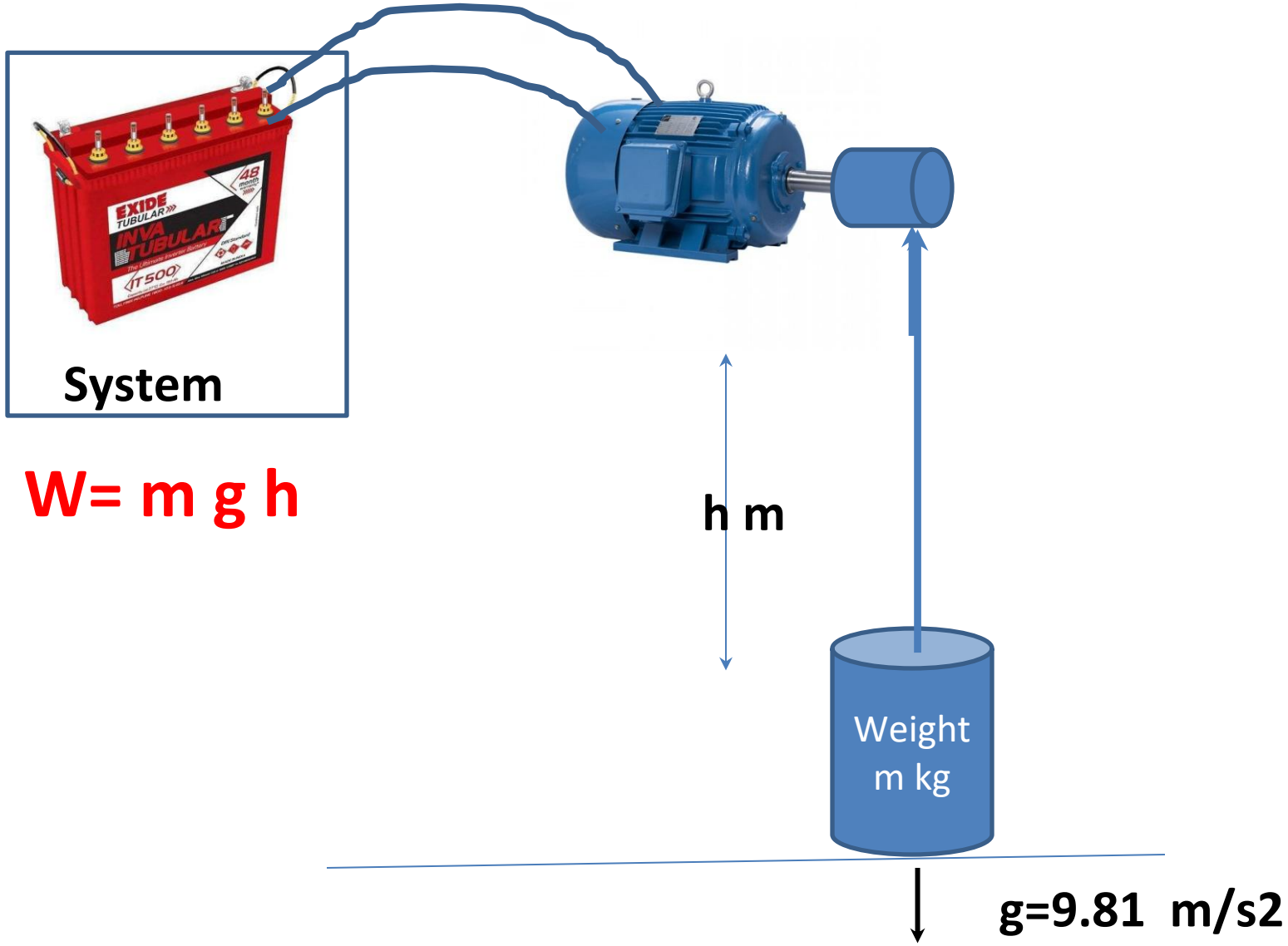
Work= Force X Displacement

W= F X L Joule/ kilo Joule

- To define all types of works in one form. Mechanical Work is defined as

- Work is said to be done by a system if the sole effect of things external to the system can be reduced to raising of a weight


System
W= m g h
Work is done by the system
h m
Weight
m kg
g=9.81 m/s2


EXIDE
TUBULAR
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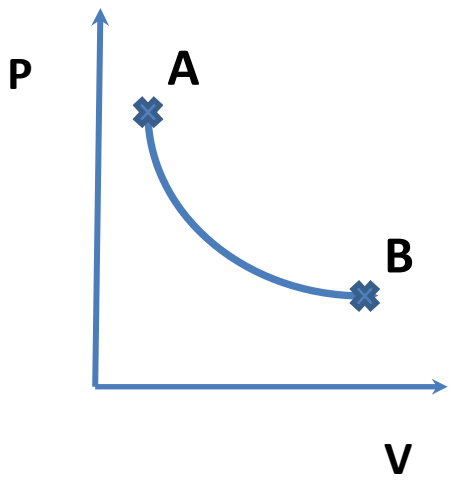
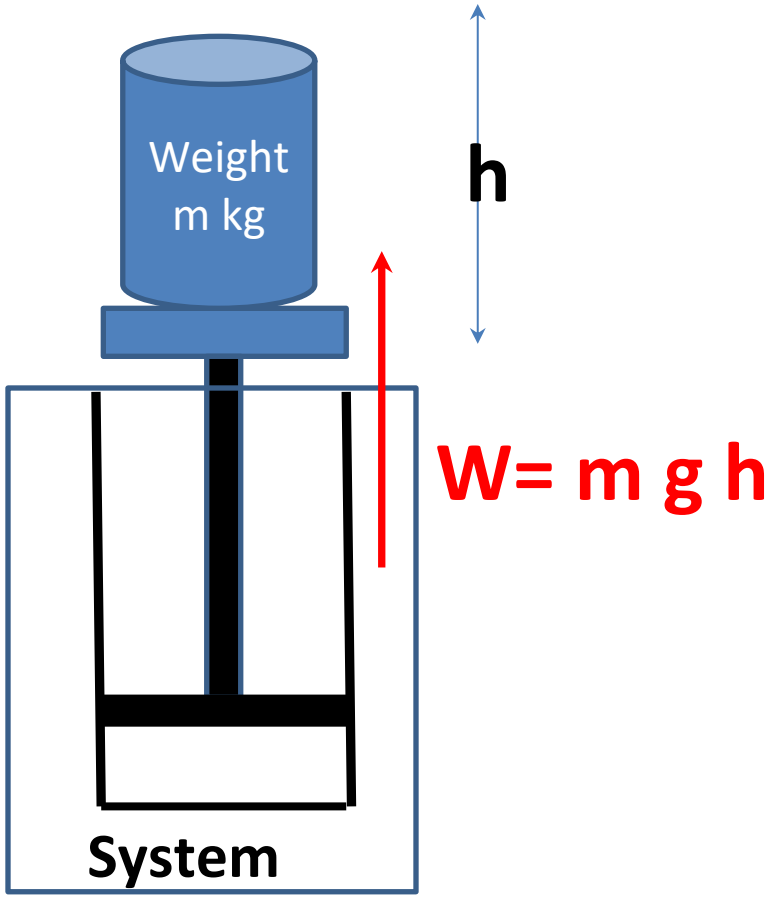
System
h m
Weight
m kg
g=9.81 m/s2

W= m g h


P
A
B
V

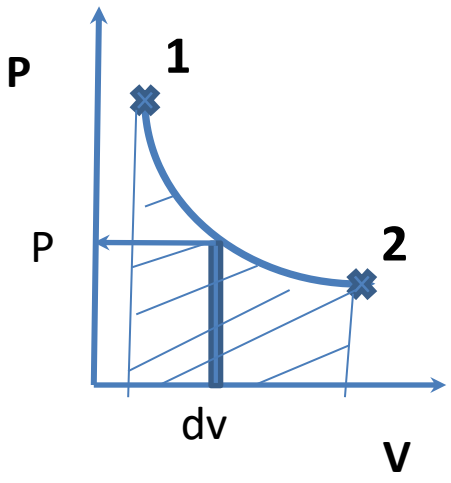

Weight
m kg
h
W= m g h
System

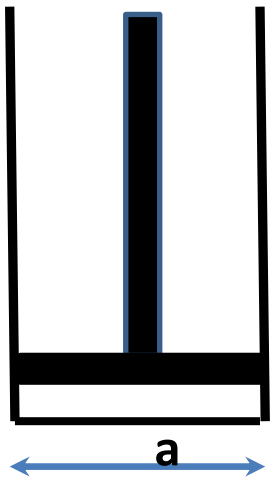
P.dV Work

$$dW = F.dl$$

$$dW = P.a.dl$$

$$dW = P.dv$$



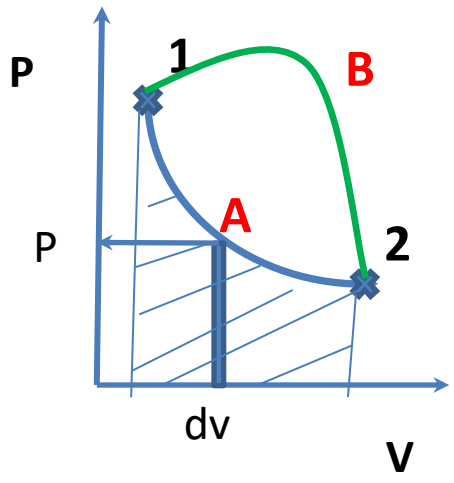


System

$$W = \int_1^2 dW = \int_1^2 P\,dv$$

$$W = \int_1^2 P\,dv$$

Work as Path Function



$$dW = P.dv$$

$$W = \int_1^2 dW = \int_1^2 P\,dv$$

$$W = \int_1^2 P\,dv$$

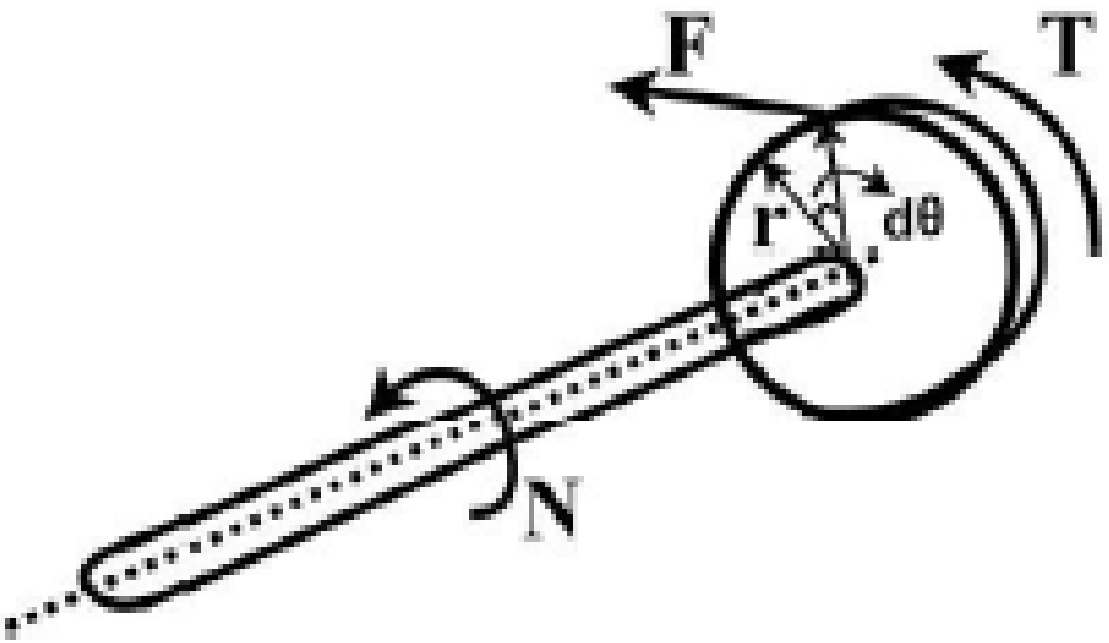
$$W_A =? W_B$$

$$\int_1^2 P.dv_{\;A} =? \int_1^2 P.dv_{\;B}$$

$$W_A \neq W_B$$

Work depends on Path

Shaft Work



$$T = F.r$$

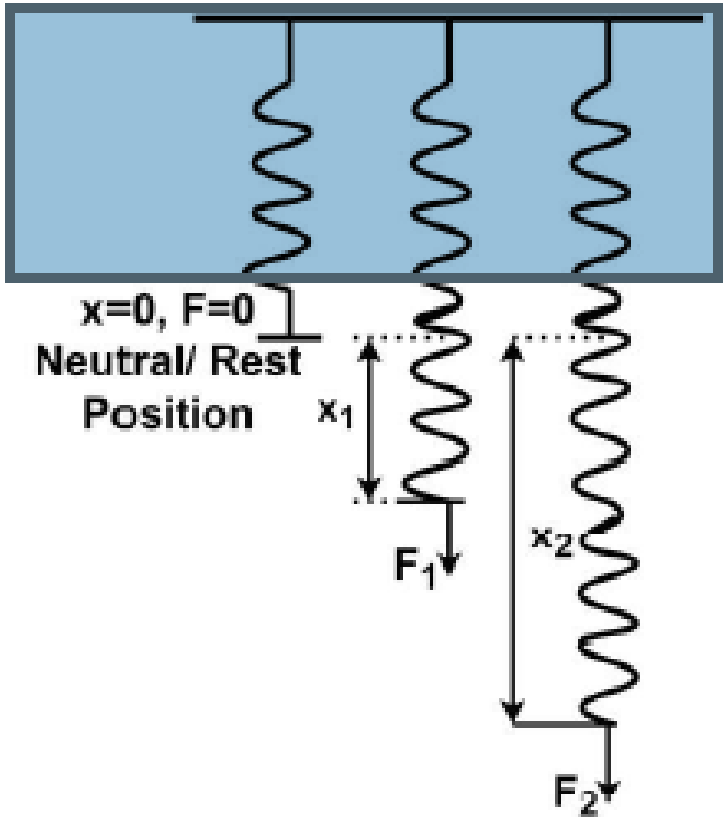
$$dw = F.rd\theta$$

$$dw = T.d\theta$$

$$\dot{W} = \frac{\delta W}{dt} = T\frac{d\theta}{dt} = T\omega$$

Here ω is the angular velocity of the shaft and is given as $\omega = \frac{2\pi N}{60}\ rad/s$ where N is the revolution of the shaft per minute.

Spring Work



$$F = kx$$

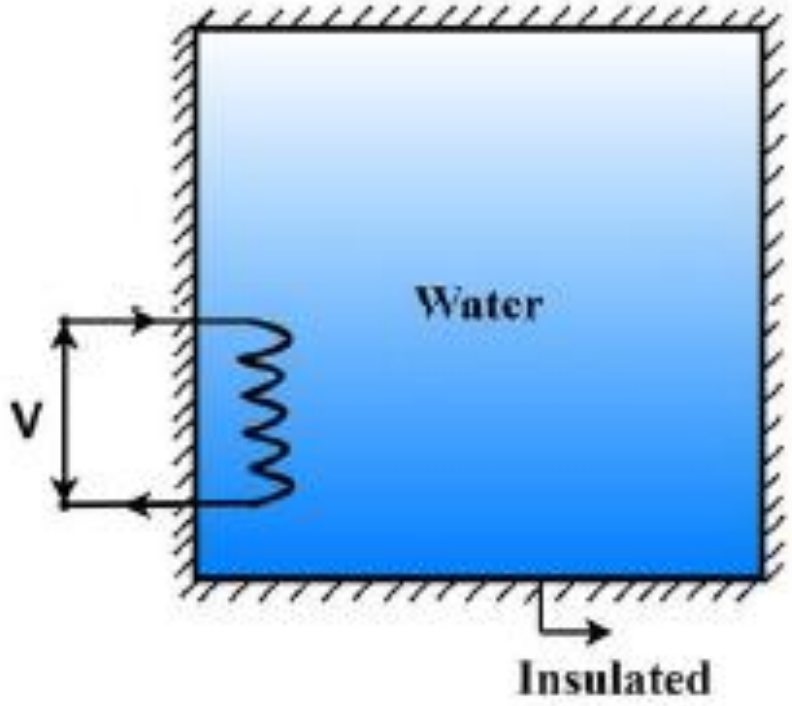
where is the spring constant and its unit is N/m

$$W = \int_{X1}^{X2} F.dx$$

$$W = \int_{X1}^{X2} kx.dx$$

$$W = \frac{1}{2}k(X2^2 - X1^2)$$

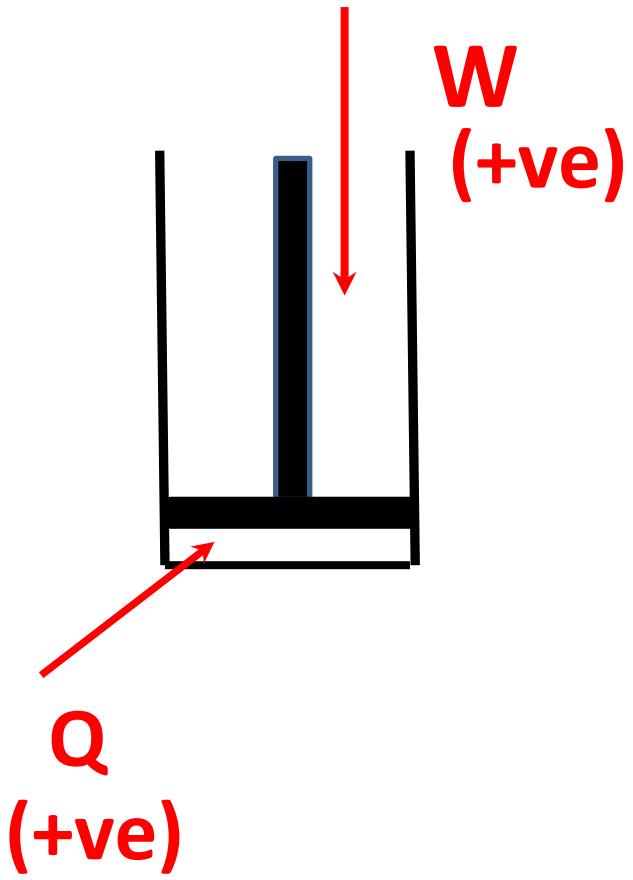
Electrical work

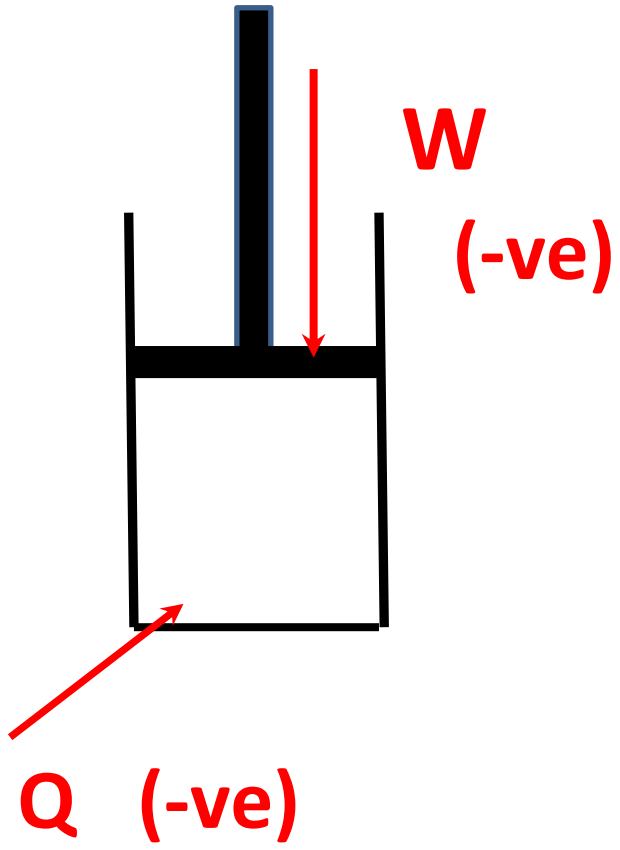


$$We = V.I$$

Here, We is called the electrical power,
the number of electrical charges moving per unit time is electric current (I) and its unit is ampere (A).

Sign Convention





Specific Heat (C)

- Amount of heat required to increase unit temperature (℃or K) of unit mass (kg) of the substance

$$C = \frac{dQ}{m.\Delta T} \qquad {}^{kJ}/_{kg.K} \quad 0r \quad {}^{kJ}/_{kg.℃}$$

$$C = \left(\frac{dq}{dT}\right)$$

$$dq = C.dT$$

Modes of heat transfer

I. Conduction

- Heat conduction is a process in which heat is transferred from the hotter part to the colder part in a body **without i**nvolving any **actual movement of the molecules** of the body.
- As a result of the vibratory motion of the molecules.
- Takes place in solids

The rate of heat transfer by conduction depends on

- Thermal conductivity
- Temperature difference
- Surface area

Fourier's law of conduction

- Heat transfer rate ($\dot{\boldsymbol{Q}}$):

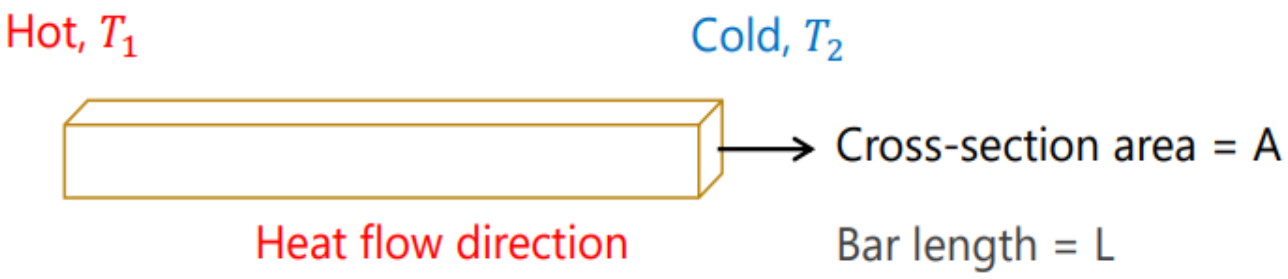
$$\dot{\boldsymbol{Q}} = -kA\frac{dT}{dx}$$

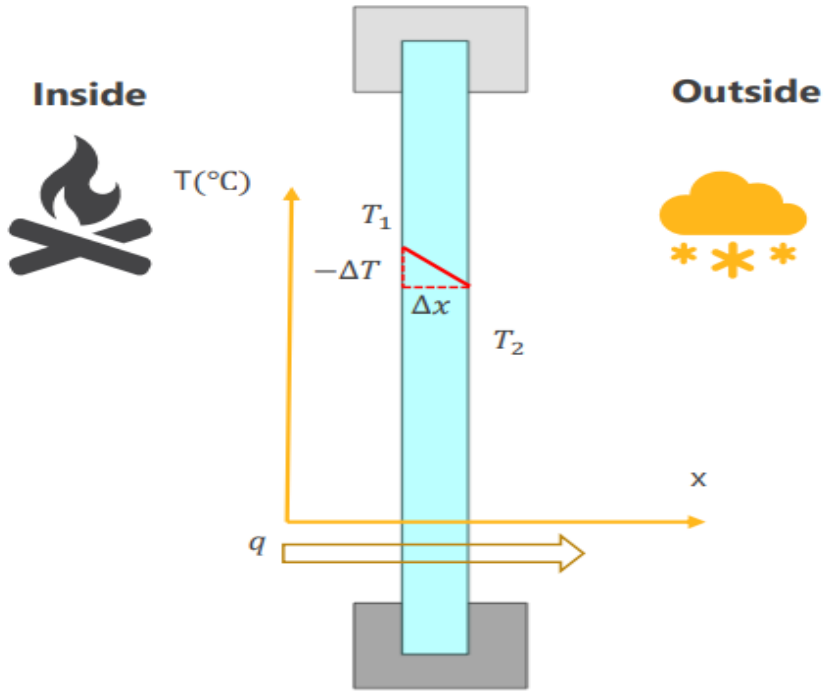
Where:

- $\dot{\boldsymbol{Q}}$= heat transfer rate (kJ/s) or (W)
- k= thermal conductivity (W/m·K)
- $\frac{dT}{dx}$ =temperature gradient
- Negative sign → heat flows from **high temperature to low temperature**

Heat flux = $\dot{\boldsymbol{Q}}$ /A (W/m^2)

$$\dot{\boldsymbol{q}} = -k\frac{dT}{dx}$$





$$\dot{q} = -k\frac{dT}{dx}$$

$$\dot{q} = -k\frac{(T_2 - T_1)}{L}$$

$$\dot{Q} = -\frac{Ak}{L}(T_2 - T_1)$$

- A metal slab of thickness **0.02 m** and area **0.5 m²** has its two faces at **100 °C** and **40 °C**. Thermal conductivity $k = 200\ W\ m^{-1}K^{-1}$. Find the **rate of heat flow**. Also find the heat transfer in 10min.

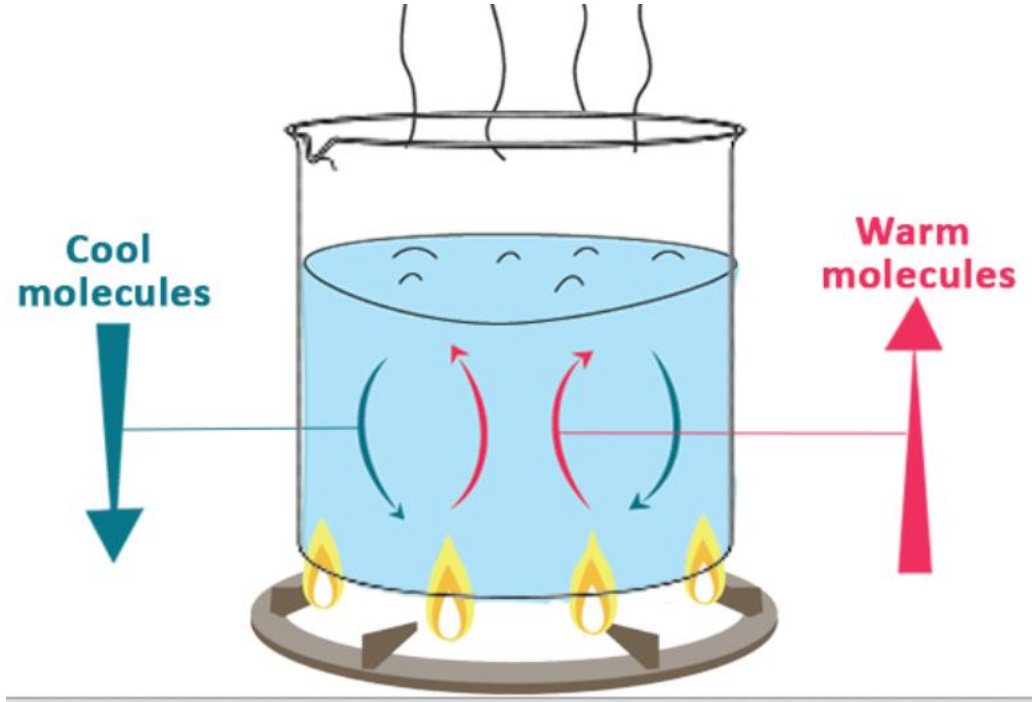
II. Convection

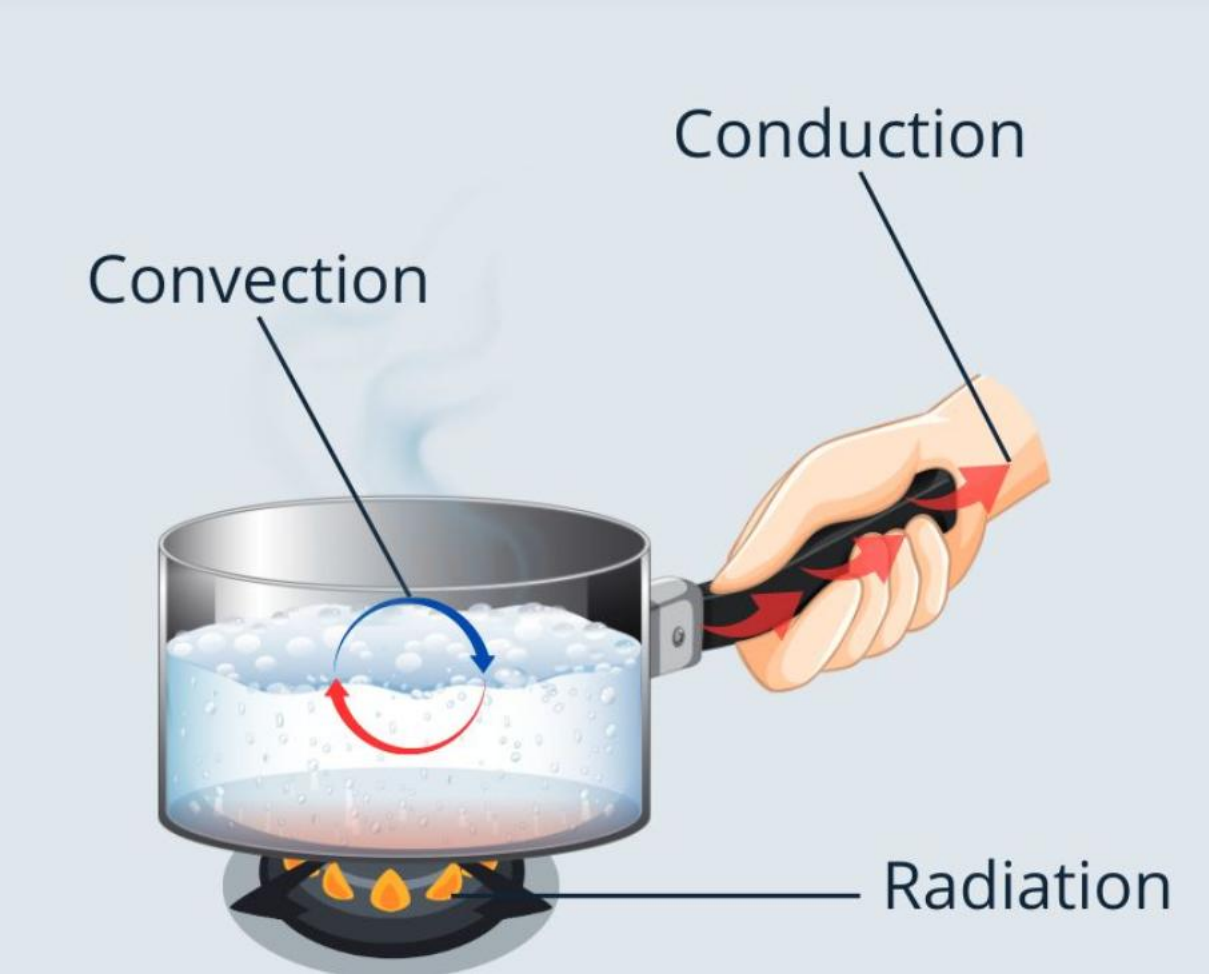
- **Convection** is the process of heat transfer by the bulk movement of molecules within fluids such as gases and liquids.

There are two types of convection:

- **Natural convection:** convection takes place due to buoyant force

Forced convection: external sources such as fans and pumps are used




Conduction
Convection
Radiation

Newton's law of cooling

- The rate at which an object cools (or warms) is **proportional to the difference between the object's temperature and the surrounding (ambient) temperature**

$$Rate\ of\ Colling \propto\ \Delta T$$

$$\frac{dT}{dt} = -k\ (\mathrm{T(t)} - \mathrm{Ta})$$

Where:

- T= temperature of the object at time t
- T_a =ambient (surrounding) temperature
- k= positive constant depending on the object and environment $k = \frac{hA_s}{mc_p}$
- $\frac{dT}{dt}$ =rate of change of temperature

1. Consider a 250 g ceramic mug of coffee initially at 90 °C, placed in a room at 20 °C. Experimental fitting yields k= 0.03 min^{-1}. Find the temperature after **10 minutes and 30 minutes**.

2. A cup of coffee at **90 °C** is placed in a room at **20 °C**. After **5 minutes**, its temperature becomes **70 °C**. Find the temperature after **10 minutes**.

- A thermometer initially at atmospheric temperature of **25 °C** is placed in water at **100 °C**. After **5 minute**, it reads **100 °C**. Find the time required to read correct human body temperature for the thermometer reading.

Newton's law of cooling

"The rate of heat loss of a body is proportional to the difference in temperatures between the body and its surroundings."

$$\dot{Q}\alpha\ (T - Ts)$$

$$\dot{Q} = \mathbf{h.A.(T(t) - Ta)}$$

Where,

$\dot{Q}$ - Heat Transfer per unit time

h- heat transfer coefficient

A - Surface Area

T- Temperature of object's surface

Ta- atmosphere temperature

T(t)= Temperature at time t

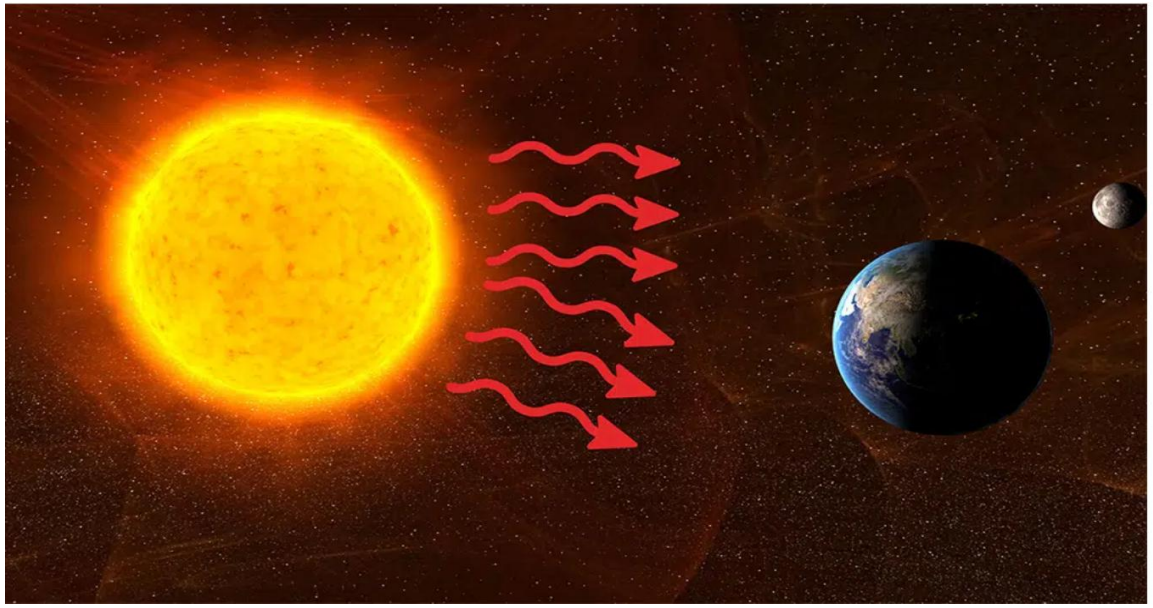
- For Heat loss rate $Q = 800\ W$, $h = 20\ W\ m^{-2}K^{-1}$, Temperature difference $= 40°C$, Find **area** A.

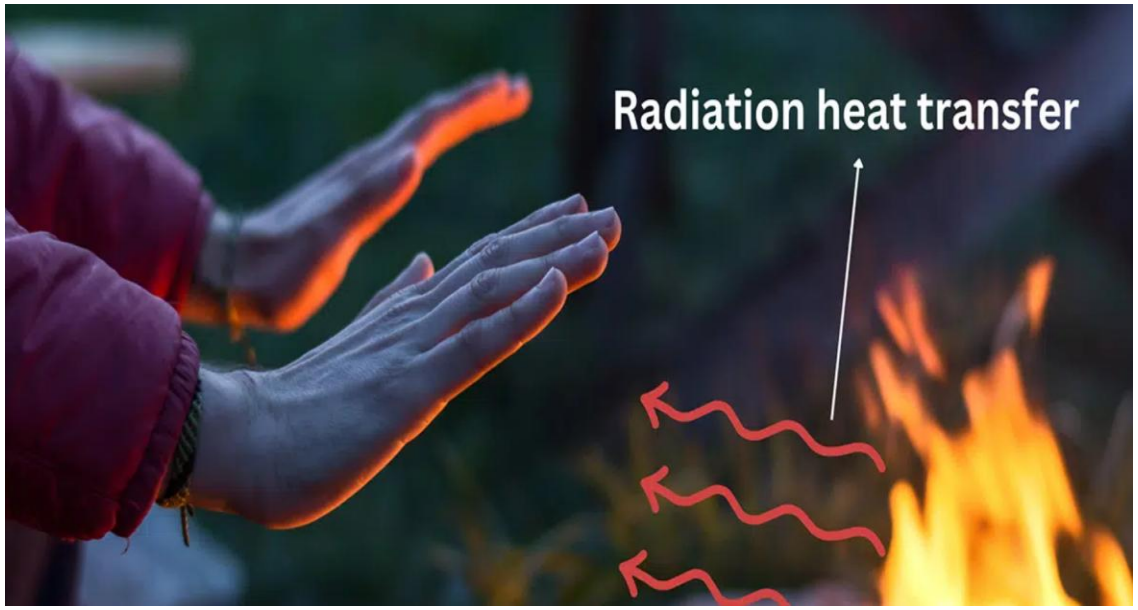
An electric heater must dissipate **2500 W** of heat to the surrounding air. Find the **minimum surface area required**. If $h = 25\ W\ m^{-2}K^{-1}$ Surface temperature $= 120°C$ Room temperature $= 20°C$

III Radiation

Radiation heat transfer is defined as, “ the transfer of energy across a system boundary by means of an electromagnetic mechanism which is caused solely by a temperature difference.

- Conduction and convection requires medium but radiation does not require medium





- The emission of the thermal radiation depends upon the nature, temperature and state of the emitting surface.

  Temperature
  Surface properties
  Distance
  The angle of incidence

- Rate of heat transfer by conduction and convection varies as temperature difference to the first power whereas radiant heat exchange between two bodies varies as temperature difference to the fourth power.

Stefan-Boltzmann law

- The thermal energy radiated by a blackbody per second per unit area is proportional to the fourth power of the absolute temperature.
- **For Black body** $$\dot{q}_{rad} \; \alpha \; T_{abs}^{4}$$
- Where, $\dot{q}$ - Heat radiator per unit area per unit time

Stefan-Boltzmann law

- **For Black body**

$$\dot{Q}_{rad} = \sigma A T_{abs}^4$$

Where, $\dot{Q}$ - Heat radiator per unit time

σ- Stefan-Boltzmann Constant = $5.69 \times 10^{-8}\ W/m^2 \cdot K^4$

A - Surface Area

- **For Real body**

$$\dot{Q}_{rad} = \varepsilon \sigma A T_{abs}^4$$

Where, ε- Emissivity of body =1 for black body

Heat Transfer by Radiation

$$\dot{Q} = F_{1-2}\varepsilon_1 A_1 \sigma\left(T_2^4 - T_1^4\right)$$

Where,

$\dot{Q}$ is the radiation <u>heat transfer rate</u> in W,

F_{1-2} is called the Shap/*view factor* between objects 1 and 2 (it describes how well object 1 "sees" object 2),

ε_1 is the dimensionless emissivity or <u>absorptivity</u> (the hotter object is said to emit energy while the colder object absorbs energy) of object 1,

A_1 is the surface area of object 1,

σ is the Stefan-Boltzmann constant (5.69×10^{-8} W/m$^2 \cdot$ K^4),

T_1 and T_2 are the absolute surface temperatures of the objects.

A hot metal plate has Area $A = 2\ m^2$, Temperature $T = 800\ K$, Emissivity $\varepsilon = 0.6$, Find the heat radiated per second

A surface has Area $A = 0.5\ m^2$, Emissivity $\varepsilon = 0.7$, Surface temperature $T = 700\ K$, Surrounding temperature $T_s = 300\ K$, Find the rate of heat loss by radiation.

Q time?

https://classroom.google.com/c/MTM2MzQxMTYxODMw